Project Documentation

Clinical and Financial Healthcare Analytics

Healthcare Data Analysis using Python

1.Project Overview

This project focuses on analyzing healthcare data to identify patterns in patient admissions, diagnoses, hospital stays, and lifestyle factors. The analysis was performed using Python and various data visualization techniques to generate meaningful insights that can support healthcare decision-making.

2. Objectives

- Understand the distribution of patient demographics.
- Analyze admission patterns and healthcare utilization.
- Explore relationships between medical and lifestyle factors.
- Identify trends and patterns through visual analytics.
- Generate actionable insights and recommendations.

3.Dataset Description

- **Dataset contains** healthcare-related information including:

- Age
- Gender
- Diagnosis
- Admission Type
- Admission Date
- Length of Stay
- Smoking Status
- Glucose Level
- Insurance Attributes
- Financial Attributes
- Other patient and hospital-related attributes

- **Dataset Source**

- Dataset Name: Healthcare Dataset
- File Name : healthcare_dataset.csv
- Source : Publicly Available Healthcare Dataset
- Dataset Size : Records-100000, Features-30

4.Tools and Technologies Used:

Tool	Purpose
Python	Programming Language
Pandas	Data Cleaning and Analysis
NumPy	Numerical Operations
Plotly	Interactive Data Visualization
Jupyter Notebook	Development Environment

5.Project Workflow

5.1 Data Collection

The healthcare dataset was obtained for educational and analytical purposes.

5.2 Data Preprocessing

- Missing value handling
- Data type correction
- Duplicate removal
- Feature inspection
- Filtering and Aggregation

5.3 Exploratory Data Analysis (EDA)

A. Univariate Analysis

- Gender Distribution
- Diagnosis Distribution
- Admission Type Distribution
- Smoking Status Distribution
- Insurance Provider Distribution
- Readmission Risk Distribution

B. Bivariate Analysis

- Average Billing Amount by Gender
- Average Length of Stay by Admission Type
- Average Glucose Level by Smoking Status
- Patient Count by Diagnosis
- Average Billing Amount by Insurance Provider

C. Multivariate Analysis

- Average Billing Amount by Gender and Admission Type
- Average Length of Stay by Smoking Status and Readmission Risk
- Average Glucose Level by Age Group and Diagnosis
- Average Billing Amount by Insurance Provider and Admission Type

D. Pivot Table

- Average Billing by Insurance Provider
- Average Billing Amount by Gender and Admission Type
- Average Cholesterol Level by Smoking Status and Readmission Risk
- Average Glucose Level by Age Group and Gender
- Average Heart Rate and Oxygen Saturation by BMI Category

E. Correlation Analysis

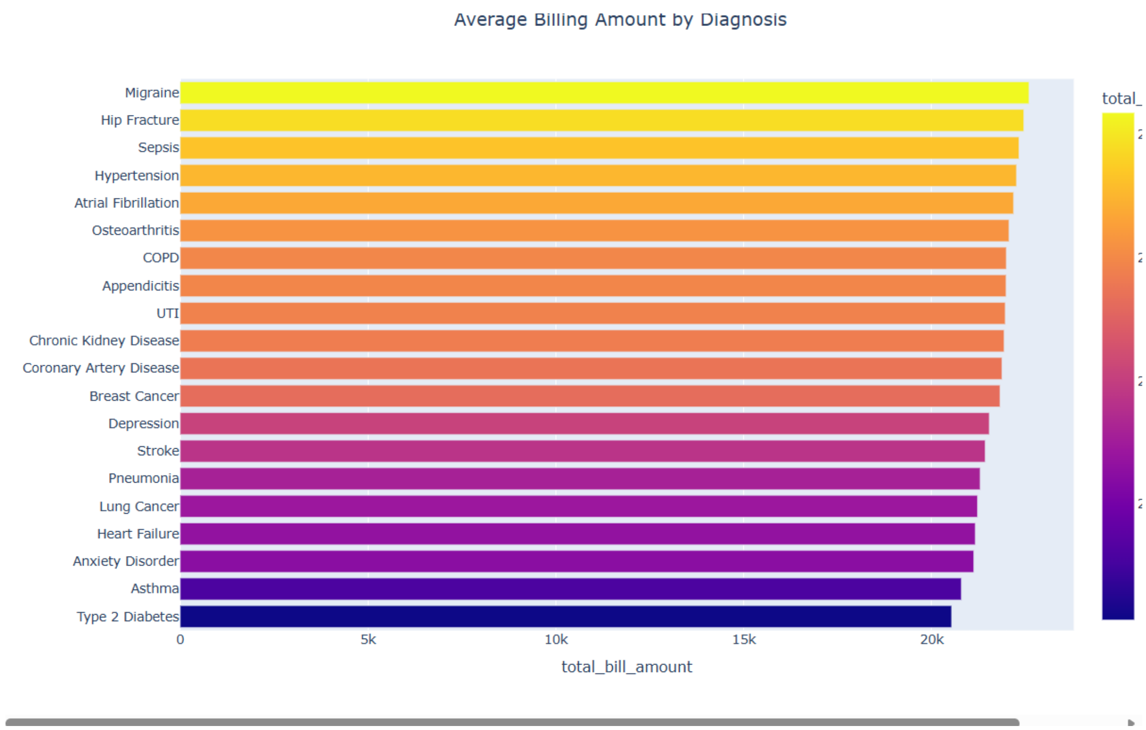
F. Statistical Summary

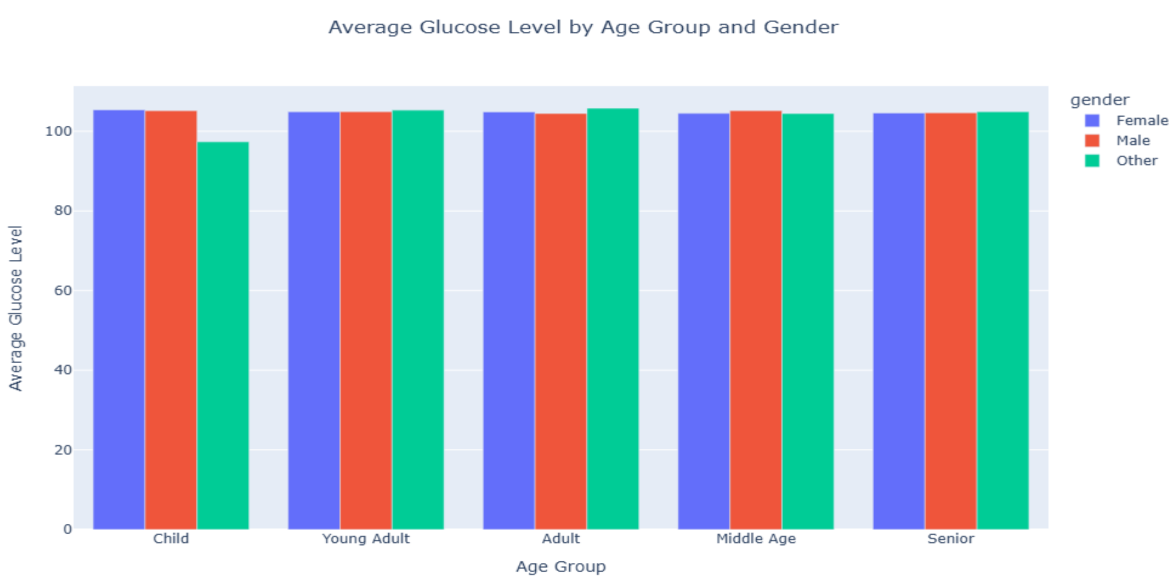
5.4 Data Visualization

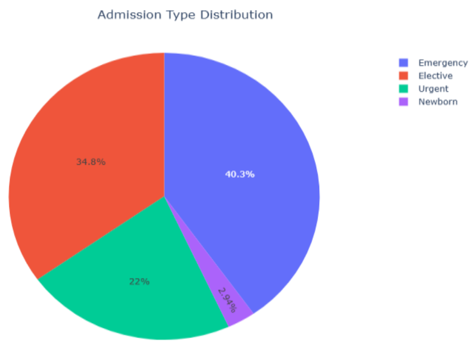
Various visualizations were created, including:

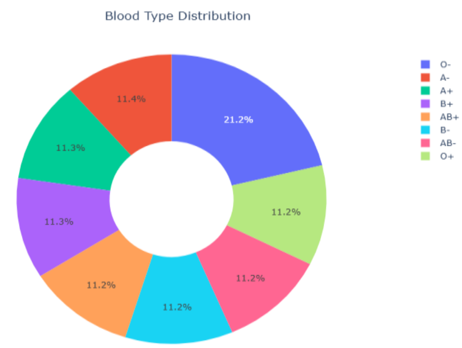
- Bar Charts
- Grouped Bar Charts
- Pie Charts
- Donut Charts
- Line Charts
- Scatter Plots
- Histogram
- Heat Maps
- Box Plots
- Tree Map
- Violin Chart
- Sunburst Chart

SCREENSHOTS OF CHARTS

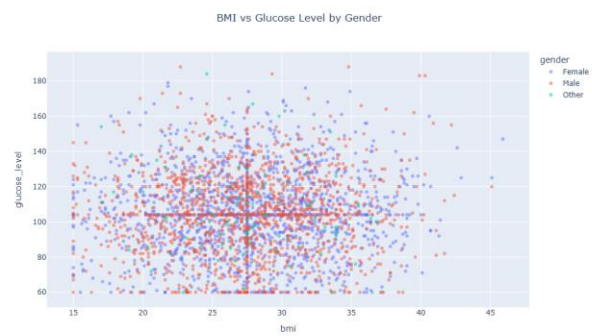


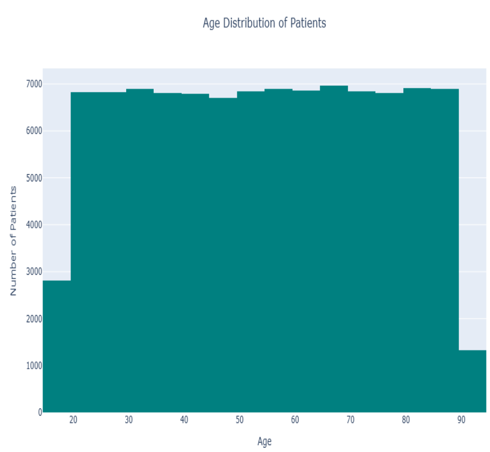


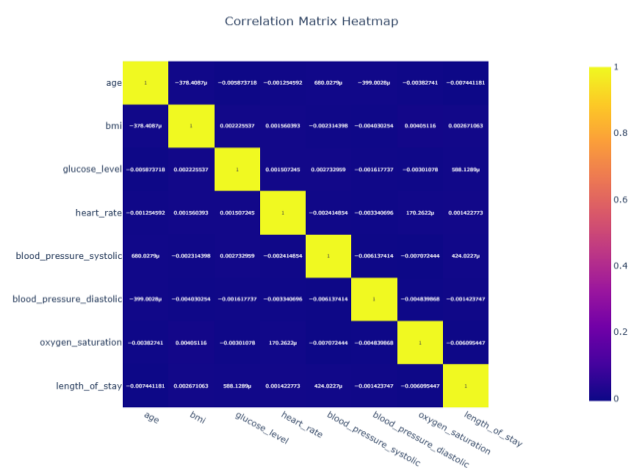

Admission Type Distribution
Emergency
Elective
Urgent
Newborn
40.3%
34.8%
22%
2.94%

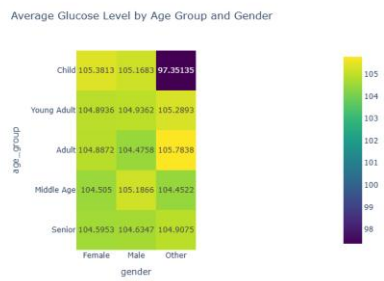

Blood Type Distribution
O-
A-
A+
B+
AB+
B-
AB-
O+
21.2%
11.4%
11.3%
11.3%
11.2%
11.2%
11.2%
11.2%

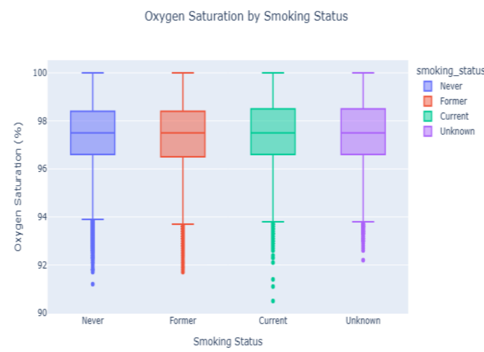

Monthly Patient Admissions Trend
Number of Admissions
8600
8400
8200
8000
7800
7600
January
February
March
April
May
June
July
August
September
October
November
December
Month

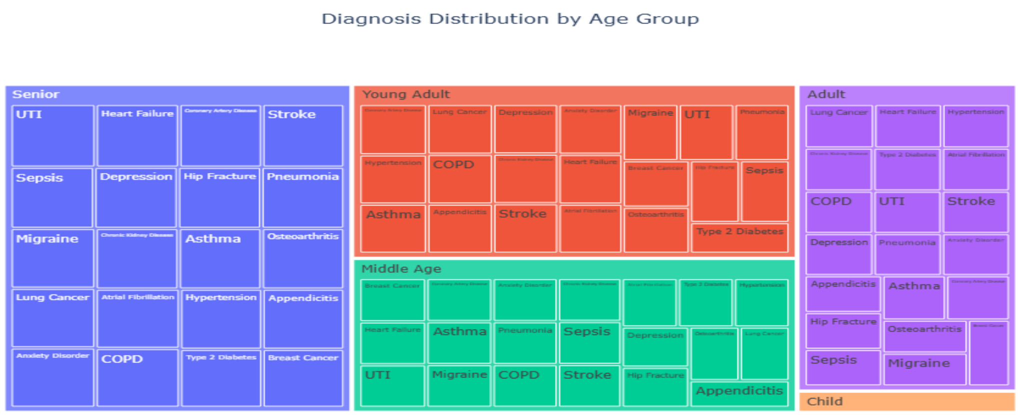

BMI vs Glucose Level by Gender
gender
Female
Male
Other
glucose_level
180
160
140
120
100
80
60
15
20
25
30
35
40
45
bmi

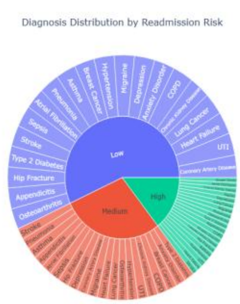

Age Distribution of Patients
Number of Patients
7000
6000
5000
4000
3000
2000
1000
0
20
30
40
50
60
70
80
90
Age

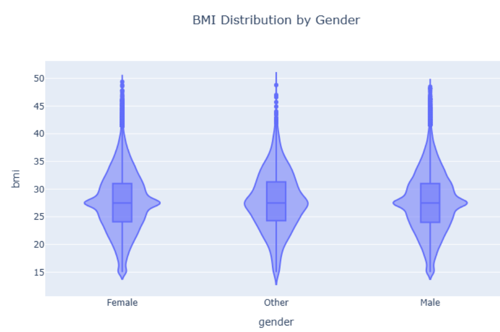

Correlation Matrix Heatmap
age
bmi
glucose_level
heart_rate
blood_pressure_systolic
blood_pressure_diastolic
oxygen_saturation
length_of_stay
1 −378.4087μ −0.005873718 −0.001254592 680.0279μ −399.0028μ −0.00382741 −0.007441181
−378.4087μ 1 0.002225537 0.001560393 −0.002314398 −0.004030254 0.00405116 0.002671063
−0.005873718 0.002225537 1 0.001507245 0.002732959 −0.001617737 −0.00301078 588.1289μ
−0.001254592 0.001560393 0.001507245 1 −0.002414854 −0.003340696 170.2622μ 0.001422773
680.0279μ −0.002314398 0.002732959 −0.002414854 1 −0.006137414 −0.007072444 424.0227μ
−399.0028μ −0.004030254 −0.001617737 −0.003340696 −0.006137414 1 −0.004839868 −0.001423747
−0.00382741 0.00405116 −0.00301078 170.2622μ −0.007072444 −0.004839868 1 −0.006095447
−0.007441181 0.002671063 588.1289μ 0.001422773 424.0227μ −0.001423747 −0.006095447 1
1
0.8
0.6
0.4
0.2
0











5.5 Key Insights

- Certain diagnoses were associated with longer hospital stays.
- Admission patterns varied across different patient groups.
- Lifestyle factors showed potential relationships with health outcomes.
- Visual analysis helped identify trends and anomalies within the dataset.

5.6 Recommendations

- Based on the findings of this study, healthcare organizations should strengthen preventive care initiatives and promote regular health screenings to identify potential health risks at an early stage.
- Patients with elevated clinical indicators such as high BMI, glucose levels, cholesterol levels, and blood pressure should be monitored closely through personalized care plans and lifestyle management programs.
- Hospitals can improve patient outcomes by implementing targeted follow-up strategies for individuals with medium and high readmission risk, thereby reducing unnecessary hospital readmissions.
- Interactive dashboards and real-time monitoring systems can also be developed to support better clinical and financial decision-making.

5.7 Future Scope

- The project can be further enhanced by incorporating machine learning and predictive analytics techniques to forecast readmission risk, patient outcomes, and healthcare costs.
- Advanced models can help identify high-risk patients at an early stage and support preventive interventions.
- Additionally, interactive dashboards and real-time monitoring systems can be developed to improve clinical decision-making, resource allocation, and overall healthcare management.

6. Files Included

The following files are attached:

- ‘raw_healthcare_dataset.csv’- Dataset used for analysis
- ‘cleaned_healthcare_dataset.csv’- Cleaned Dataset
- ‘Healthcare_Analytics_Project.ipynb’- Jupyter Notebook Source File
- README.md- Project description and documentation

7. How to Use

- Open ‘raw_healthcare_dataset.csv’ file to view the raw dataset.
- Open ‘cleaned_healthcare_dataset.csv’ to view the cleaned dataset
- Open the Jupyter Notebook Source File ‘Healthcare_Analytics_Project.ipynb’ to view the data analysis and explore the interactive visuals.

Conclusion

The project successfully demonstrates how healthcare data can be analyzed using Python-based data analytics techniques. The findings provide valuable insights into patient characteristics, healthcare utilization patterns, and financial aspects of healthcare services.

Author

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